Assignment No	**3**
Title	**Static & Non Static problems**
Objective	**1. Employee Salary using Non Static Keyword** **2. Non Static program(Counter)** **3. Static Program(Counter)** **4. Student Program(Static)** **5. Employee Salary using Static keyword**
Roll No	**MCA2649**

1) Employee Salary using Non Static Keyword

Source Code:-

Without Static

```
import java.util.*;
class Employee
{
        //Instance Variable
      String name;
        int id;
        float salary;
        float income;

        //Constructor
        Employee(String name,int id,float salary)
        {
```

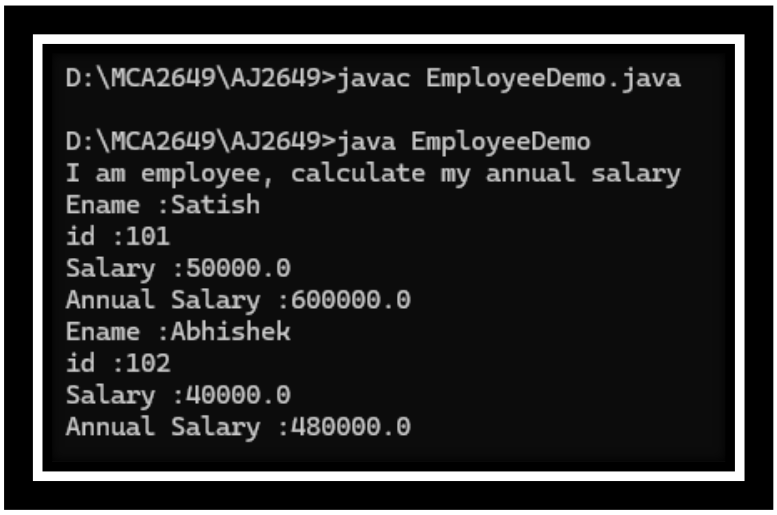
```
            this.name = name;
            this.id = id;
            this.salary = salary;
        }

        //Method 1 Logic
        void calcSalary()
        {
            income = this.salary*12;
        }

        //Method 2 Print
        public void print()
        {
            System.out.println("Ename :" + name);
            System.out.println("id :" + id);
            System.out.println("Salary :" + salary);
            System.out.println("Annual Salary :" + income);
        }
}
class EmployeeDemo
{
        public static void main(String args[])
        {
            System.out.println("I am employee, calculate my annual
salary");
```

```
            Employee e1 = new Employee("Satish",101,50000);
            Employee e2 = new Employee("Abhishek",102,40000);
          e1.calcSalary();
            e1.print();
            e2.calcSalary();
            e2.print();
        }
}
```

Output:-



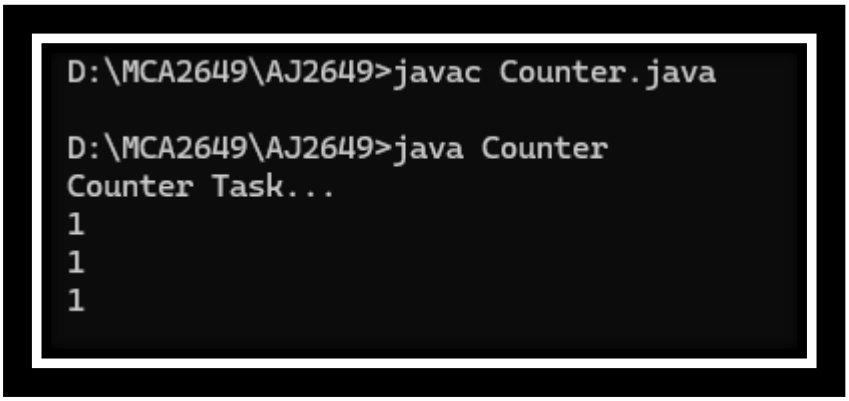
2)Non Static program(Counter)

Source Code:-

```
class Counter
{
        // instance variable
        int count=0;
        Counter()
```

```
        {
                count++;
                System.out.println(count);
        }
        public static void main(String args[])
        {
                System.out.println("Counter Task...");
                Counter c1 = new Counter();
                Counter c2 = new Counter();
                Counter c3 = new Counter();
        }
}
```

Output:-



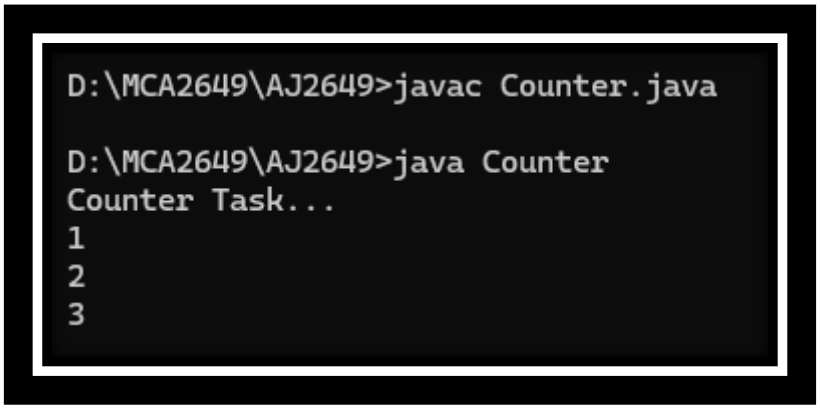
3) Static Program(Counter)

Source Code:-

```
class Counter
{
        // instance variable
```

```
    static int count=0;
    Counter()
    {
        count++;
        System.out.println(count);
    }
    public static void main(String args[])
    {
        System.out.println("Counter Task...");
        Counter c1 = new Counter();
        Counter c2 = new Counter();
        Counter c3 = new Counter();
    }
}
```

Output:-



4) Student Program(Static)

Source Code:-

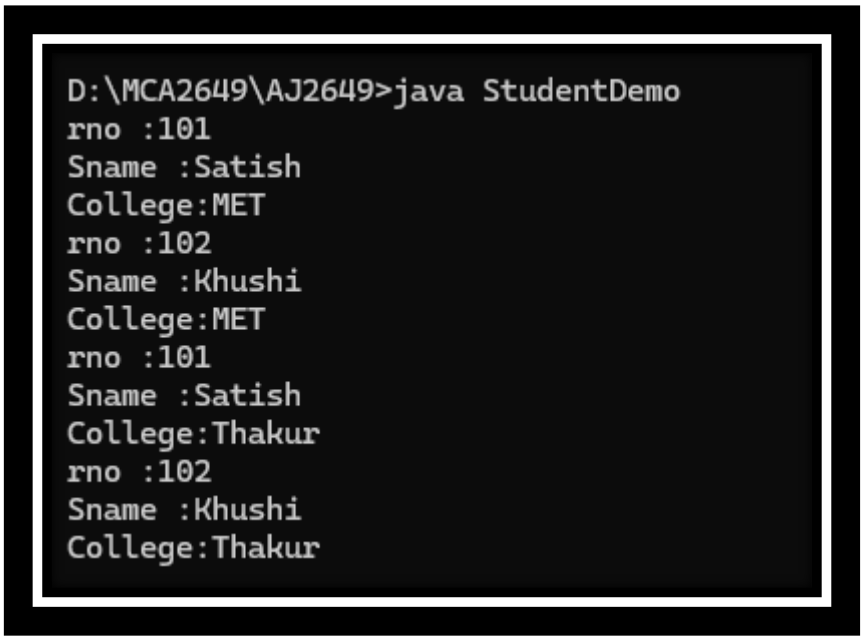
```
class Student
{
        //Instance Variable
        int rno;
      String name;
        static String college = "MET";

        //Constructor
        Student(int rno,String name)
        {
                this.rno = rno;
                this.name = name;
        }

        //Method 1 Print
        public void print()
        {
                System.out.println("rno :" + rno);
                System.out.println("Sname :" + name);
                System.out.println("College:" + college);
        }
}
class StudentDemo
{
        public static void main(String args[])
```

```
    {
        Student s1 = new Student(101,"Satish");
        Student s2 = new Student(102,"Khushi");
        s1.print();
        s2.print();
        Student.college = "Thakur";
        s1.print();
        s2.print();
    }
}
```

Output:-



5) Employee Salary using Static keyword

Source Code:-

```
class Employee
{
        // Instance variables
        String name;
        int id;
        float salary;
        float income;

        static int nextId = 1;

        // Constructor
        Employee(String name, float salary)
        {
        this.name = name;
        this.id = nextId++;
        this.salary = salary;
        }

        // Calculate annual salary
        void calcSalary()
        {
        income = salary * 12;
        }

        // Print employee details
        public void print()
        {
        System.out.println("Ename : " + name);
        System.out.println("id : " + id);
        System.out.println("Salary : " + salary);
        System.out.println("Annual Salary : " + income);
        }
}

class EmployeeDemoo
{
```

```
        public static void main(String args[])
        {
        System.out.println("I am employee, calculate my annual
salary");

        Employee e1 = new Employee("Satish", 50000);
        Employee e2 = new Employee("Abhishek", 40000);

        e1.calcSalary();
        e1.print();

        e2.calcSalary();
        e2.print();
        }
}
```

Output:-

```
D:\MCA2649\AJ2649>javac EmployeeDemoo.java

D:\MCA2649\AJ2649>java EmployeeDemoo
I am employee, calculate my annual salary
Ename : Satish
id : 1
Salary : 50000.0
Annual Salary : 600000.0

Ename : Abhishek
id : 2
Salary : 40000.0
Annual Salary : 480000.0
```